

Question	Marking Guidance	Mark	Comments
01.1	Any two of the following; Concentration of enzyme Volume of substrate solution pH	1	Allow same concentration of substrate
01.2	Ratio between 5.18:1 and 5.2:1; Initial rates incorrect but correctly used = 1 mark	2	Allow 1 mark if rate at: $60\text{ }^{\circ}\text{C} = 0.83\text{ g dm}^{-3}\text{ s}^{-1} / 49.8\text{ g dm}^{-3}\text{ minute}^{-1}$ OR $37\text{ }^{\circ}\text{C} = 0.16\text{ g dm}^{-3}\text{ s}^{-1} / 9.6\text{ g dm}^{-3}\text{ minute}^{-1}$
01.3	At $60\text{ }^{\circ}\text{C}$: 1. More kinetic energy; 2. More E–S complexes formed;	2	Allow converse for $37\text{ }^{\circ}\text{C}$
01.4	Different times: 1. Higher temperature / $60\text{ }^{\circ}\text{C}$ causes denaturation of all of enzyme; 2. Reaction stops (sooner) because shape of active site changed; Different concentrations of product (at $60\text{ }^{\circ}\text{C}$) 3. Substrate still available (when enzyme denatured); 4. But not converted to product;	4	Accept converse for $37\text{ }^{\circ}\text{C}$ 2. Reject if active site on substrate
